T(n) = 2T(n/3) + n log n

2T(n/3) = 2^2 T(n/3^2) + 2 * n/3 log n/3

..

2^k T(n/3^k) = 2^(k+1) T(n/3^(k+1)) + n * (2/3)^k * log (n/3^k)

k+1 = log3 n

T(n) = n^(log 3 2) * T(1) + sum n * (2/3)^i * (log n – i * log 3), i=0:k

T(n) >= n log n

T(n) <= n^(log3 2) * T(1) + n * log n * 3 – n * log 3 * sum (2/3)^i * i, i=0:k

S = sum i * a^i, i=1:n

S = a + 2* a^2 + 3* a^3 +... n* a^n

a* S = a^2 + 2* a^3 + ..(n-1)*a^n + n * a^(n+1)

S(1-a) = a + a^2 + ... + a^n – n* (a^n+1) = a(1-a^n)/(1-a) – n* a^(n+1)

S = a(1-a^n)/(1-a)^2 – n* a^(n+1)/(1-a)

a < 1:

S-> a/(1-a)^2

a=2/3 => S -> 2/3/(1-2/3)^2 = 9*2/3 = 6

=> T(n) <= n^*log3 2* Theta(1) + n log n * 3 – n log 3 * 6 =>

T(n) <= Theta(n log n)

=> T(n) = Theta(n log n)

T(n) = 4T(n/2) + n^2/log n

4T(n/2) = 4^2 T(n/2^2) + (n^2/4) *(4 / log(n/2))

4^2 T(n/2^2) = 4^3 T(n/2^3) + n^2/ 4^2 * 4^2 /log (n/2^2)
...
4^kT(n/2^k) = 4^(k+1)T(n/2^(k+1)) + n^2/ 4^k * 4^k / log(n/2^k)

k+1 = log n

T(n) = n^2* Theta(1) + n^2 * sum 1/ (log n -i), i=0:k

sum 1/(log n – i), i=0:k

1/log n + 1/(log n -1) + 1/ (log n -2) +... 1/1 =

1/1 + 1/2 +... + 1/ log n

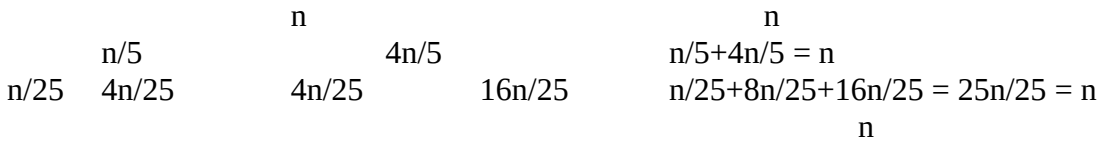
sum 1/i ~= ln (log n) = log (log n) / ln e
i=1:log n

sum 1/i^k -> convergenta
k>1

T(n) = n^2 * Theta(1) + n^2 * log (log n)/ ln e

T(n) = Theta(n^2 * log (log n))

T(n) = T(n/5) + T(4n/5) + n



numar minim de nivele:

n/5^k = 1 = > 5^k = log n => k = log5 n => T(n) >= log5 n * n = n* log n/log 5 = Theta(n log n)

numar maxim de nivele:

n/(5/4)^k = 1=> k = log5/4 n => T(n) <= log5/4 n * n = n * log n/log 5/4 = Theta(n log n)

T(n) = Theta(n log n)

T(n) = T(n/2) + T(n/4) + n^2

		n		n^2
	n/2	n/4		n^2/4 + n^2/ 16 = n^2 * 5/16
n/4	n/8	n/8	n/16	n^2 * (5/16)^2
..				n^2 * (5/16)^k

T(n) >= n^2

T(n) <= n^2 (1 + 5/16 + (5/16)^2 + ...) = n^2 * 16/11

T(n) = Theta(n^2)

T(n) = T(n/2 + log n) + n

T(n) >= n

log n <= n/4, pentru orice n >= n0 = 16

T(n) <= T(n/2 + n/4) + n = T(3n/4) + n

T'(n) = T'(3n/4) + n => master, cazul 3

n*3/4 <= c *n, pentru c = 3/4

T'(n) = Theta(n) => T(n) <= Theta(n)

=> T(n) = Theta(n)